

IMAT QUESTION BANK: Chemistry (Medium-Hard)

65 questions · 108 minutes · scoring: +1.5 correct, -0.4 wrong, 0 blank · Question bank · answer + worked solution follow every question

Each question is immediately followed by its answer and worked solution, so this paper is for review and paper study rather than blind simulation. Letters refer to the option order printed here; the online app shuffles options on every attempt.

Section A - Organic chemistry and isomerism (Q1-13)

1. D-glucose and D-galactose are both aldohexoses with the same bonding order; their Fischer projections differ only at carbon 4, where the OH points right in glucose and left in galactose. Which term most precisely classifies the pair?

- A) They are enantiomers, because the OH at C-4 is on opposite sides in the two sugars.
- B) They are identical compounds drawn as different Fischer projections.
- C) They are epimers, because they differ in configuration at exactly one chiral center, C-4.
- D) They are constitutional isomers, because in galactose the C-4 OH is bonded to a different carbon.
- E) They are anomers, because any two aldohexoses that differ at a single carbon are anomers.

Answer: C

Both sugars share the same molecular formula and bonding order, and their Fischer projections differ at exactly one stereocenter, C-4. Sugars of identical connectivity that differ in configuration at a single chiral center are epimers by definition. The mirror-image claim fails because enantiomers require every stereocenter to be inverted, while C-2, C-3 and C-5 still match, and the OH is bonded to C-4 in both sugars, so no connectivity has changed.

2. A synthesis produces a molecule with the same bonding order as (2R,3S,4R)-hexane-2,3,4-triol but with configurations (2S,3R,4S) at its three chiral centers. Which term most precisely classifies the relationship between the two molecules?

- A) They are constitutional isomers, because the OH groups occupy different positions.
- B) They are epimers, because several chiral centers change at the same time.
- C) They are diastereomers, because some of the stereocenters are inverted.
- D) They are enantiomers, because every chiral center is inverted, giving two non-superimposable mirror images.
- E) They are identical molecules, because they share the same molecular formula.

Answer: D

The second molecule inverts all three stereocenters of (2R,3S,4R) to give (2S,3R,4S), so it is the exact mirror image of the first, and two non-superimposable mirror images are enantiomers. Diastereomers must keep at least one stereocenter unchanged, and epimers differ at exactly one, so neither label fits. The bonding order never changed, so they are not constitutional isomers either.

3. In water, D-glucose cyclises and exists as two forms that differ only at carbon 1: in the alpha form the new OH points down from the ring, in the beta form it points up, while the configurations at carbons 2 to 5 are identical. Which term most precisely classifies the two forms?

- A) Conformational isomers, interconvertible by rotation about a single bond.
- B) Anomers, because they differ in configuration at the anomeric carbon, C-1.
- C) Constitutional isomers, because the OH is attached to the ring in two different ways.
- D) Enantiomers, because the anomeric carbon is the most important stereocenter.
- E) Identical compounds, because the two forms have the same molecular formula.

Answer: B

The two cyclic forms differ only at C-1, the anomeric carbon created when the aldehyde closes into a hemiacetal ring, and sugars differing in configuration at the anomeric carbon are anomers by definition. They are not conformers, because switching alpha to beta requires breaking and re-forming the C-1 bond rather than rotating about it, and they are not mirror images because the configurations at C-2 to C-5 are identical.

4. Butane-2,3-diol exists in stereoisomeric forms: one has the configuration (2R,3R) and another has (2R,3S). Which term most precisely classifies these two forms?

- A) They are conformational isomers, because they interconvert by bond rotation at room temperature.
- B) They are constitutional isomers, because an OH group has moved from C-2 to C-3.
- C) They are enantiomers, because one form contains an R descriptor and the other an S descriptor.
- D) They are identical, because rotation about the C-2 to C-3 bond superimposes them.
- E) They are diastereomers, because they are stereoisomers that are not mirror images of each other.

Answer: E

The mirror image of (2R,3R) is (2S,3S), so the (2R,3S) form is not the mirror image of (2R,3R): the pair are stereoisomers that are not mirror images, which is the definition of diastereomers. Keeping one center unchanged at R is exactly what rules out the enantiomer description, and no bond rotation can interconvert them, because rotation about the C2-C3 bond never changes a configuration.

5. Compound X is $\text{CH}_3\text{OCH}_2\text{CH}_3$ (methoxyethane) and compound Y is $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ (propan-1-ol); both have molecular formula $\text{C}_3\text{H}_8\text{O}$. Which statement is correct?

- A) They are constitutional isomers with different functional groups: an ether and an alcohol.
- B) They are enantiomers, because the alcohol is a chiral molecule.
- C) They are diastereomers, because their physical properties differ.
- D) They are the same compound drawn differently, since both have formula $\text{C}_3\text{H}_8\text{O}$.
- E) They are both ethers, because both contain a C-O single bond.

Answer: A

Both compounds have the formula $\text{C}_3\text{H}_8\text{O}$, but methoxyethane links its oxygen between two carbons (a C-O-C ether link) while propan-1-ol bonds its oxygen to hydrogen (a C-O-H alcohol group), so the atoms are connected differently: they are constitutional isomers of the functional-group type. The enantiomer claim fails because propan-1-ol has no chiral center, since its middle carbon carries two identical hydrogen atoms.

6. D-glucose is an aldohexose with an aldehyde group at C-1, while D-fructose is a ketohexose with a ketone group at C-2. Both have molecular formula $C_6H_{12}O_6$. Which term most precisely classifies the pair?

- A) Epimers, because they differ at one carbon atom.
- B) Stereoisomers, because only the position of the carbonyl differs.
- C) Constitutional isomers, because the same atoms are joined in a different order.
- D) Anomers, because the carbonyl sits at C-2 in one sugar and C-1 in the other.
- E) Enantiomers, because both are hexoses with the same formula.

Answer: C

Glucose and fructose share the formula $C_6H_{12}O_6$, but the carbonyl oxygen is joined to C-1 in the aldehyde and to C-2 in the ketone, so the same atoms are connected in a different order: they are constitutional isomers. Epimers and anomers must keep identical connectivity and differ only at a stereocenter, which is not the case, and moving a carbonyl is a connectivity change, not a stereochemical one.

7. The structure $CH_3CH_2CONH_2$ is named:

- A) Propanamide
- B) Propanenitrile
- C) Propanoic acid
- D) Propanamine
- E) Methyl ethanoate

Answer: A

A carbonyl carbon directly bonded to NH_2 is the amide group, and the three-carbon chain CH_3CH_2CO- gives the name propanamide. Propanamine would need an NH_2 attached to a saturated carbon with no $C=O$, methyl ethanoate is the ester CH_3COOCH_3 , and propanenitrile would carry a $C\equiv N$ group instead.

8. Oil of wintergreen (methyl 2-hydroxybenzoate) consists of a benzene ring bearing an OH group attached directly to the ring, plus a $COOCH_3$ group joined through the carbonyl carbon. Which two functional groups does the molecule contain?

- A) A phenol and an amide
- B) An ether and a ketone
- C) A carboxylic acid and an ester
- D) A phenol and an ester
- E) An alcohol and a carboxylic acid

Answer: D

An OH bonded directly to a benzene ring is a phenol rather than an alcohol, and the $COOCH_3$ linkage (carbonyl carbon joined through oxygen to CH_3) is an ester. No free $COOH$ remains, because the parent carboxylic acid group has been converted into the ester, so the molecule contains a phenol and an ester, not an alcohol or a carboxylic acid.

9. The structure $CH_3CH_2OCH_2CH_3$ is named:

- A) Butanal
- B) Ethoxyethane
- C) Butan-1-ol
- D) Butan-2-one
- E) Butanoic acid

Answer: B

One oxygen atom bridges two CH_2CH_3 groups, the defining C-O-C linkage of an ether, so the name is ethoxyethane, commonly called diethyl ether. Butan-1-ol shares the formula $C_4H_{10}O$ but has a terminal OH on a carbon chain, and butanal, butan-2-one and butanoic acid all require a carbonyl carbon, which this structure lacks.

10. Which of the following compounds is a ketone and contains exactly four carbon atoms?

- A)** Butanal, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$
- B)** Propanone, CH_3COCH_3
- C)** Butan-2-ol, $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$
- D)** Butanoic acid, $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$
- E)** Butan-2-one, $\text{CH}_3\text{COCH}_2\text{CH}_3$

Answer: E

A ketone has its $\text{C}=\text{O}$ on an internal carbon of the chain. Butan-2-one, $\text{CH}_3\text{COCH}_2\text{CH}_3$, meets both requirements: a four-carbon chain and an internal carbonyl. Butanal and butanoic acid reach the carbon count but carry an aldehyde and a carboxylic acid group, propanone is a true ketone with only three carbons, and butan-2-ol is an alcohol with no $\text{C}=\text{O}$ at all.

11. An aldohexose is drawn as a Fischer projection with the aldehyde at the top as C-1. The OH groups at C-2, C-3 and C-4 point right, right and left respectively, and the OH at C-5, the chiral center farthest from the aldehyde, points left. How is the sugar classified?

- A)** It is an L-sugar, because the OH on the stereocenter farthest from the aldehyde points left.
- B)** It is meso, because left- and right-pointing OH groups balance two against two.
- C)** It is a D-sugar, because the aldehyde group is drawn at the top of the projection.
- D)** It is a D-sugar, because D/L is fixed by the OH closest to the aldehyde group.
- E)** It is a D-sugar, because two of its four OH groups point right.

Answer: A

The D or L label is decided solely by the chiral center farthest from the carbonyl group, C-5 in an aldohexose: a left-pointing OH there places the sugar in the L series. What happens at C-2 to C-4, including any two-right, two-left balance of the OH groups, is irrelevant to the label, and meso describes internal mirror symmetry, not the D/L classification.

12. Maltose and cellobiose are both disaccharides of two glucose units with molecular formula $\text{C}_{12}\text{H}_{22}\text{O}_{11}$. Maltose joins its units with an α -1,4 glycosidic bond and cellobiose with a β -1,4 bond. Human enzymes hydrolyse maltose efficiently but cannot digest cellobiose. Which statement explains this best?

- A)** Maltose is digested because it is a reducing sugar, whereas cellobiose is non-reducing.
- B)** Cellobiose passes through the gut unchanged because no enzyme can release glucose from a β bond.
- C)** Cellobiose cannot be digested because its molecular formula differs from that of maltose.
- D)** Human enzymes fit the α -1,4 geometry of maltose but not the β -1,4 geometry of cellobiose, so cellobiose is not hydrolysed.
- E)** Cellobiose is indigestible because β glycosidic bonds are chemically stronger than α bonds and cannot be broken in water.

Answer: D

The two disaccharides contain exactly the same atoms; only the three-dimensional geometry of the link differs, α -1,4 in maltose versus β -1,4 in cellobiose. Enzyme active sites are stereochemically specific, so the human enzyme fits the α geometry and hydrolyses maltose but cannot bind cellobiose. The claim that β bonds are chemically stronger fails because both bond types hydrolyse given the right catalyst (human lactase even digests the β bond of lactose), and cellobiose, with one free anomeric carbon, is itself a reducing sugar.

13. In sucrose, the glycosidic bond links the anomeric carbon C-1 of glucose to the anomeric carbon C-2 of fructose. In maltose, the glycosidic bond uses the anomeric carbon of only one of its two glucose units. Which statement is correct?

- A)** Sucrose is reducing because fructose is a ketose, and ketoses are always stronger reducing agents.
- B)** Sucrose is non-reducing because both anomeric carbons are locked inside its glycosidic bond, while maltose is reducing because one anomeric carbon stays free.
- C)** Maltose is non-reducing, because it is built only from glucose units.
- D)** Sucrose is reducing, because the fructose ring can still open at C-2 while glycosidically bonded.
- E)** Both sugars are non-reducing, because any glycosidic bond blocks reduction.

Answer: B

A sugar is reducing only if it has a free anomeric carbon that can open to the carbonyl form. Sucrose locks both anomeric carbons (C-1 of glucose and C-2 of fructose) inside its glycosidic bond, so neither ring can open and sucrose is non-reducing, while maltose leaves the second glucose's anomeric carbon free and is reducing. The claim that any glycosidic bond blocks reduction fails because only the anomeric centers taking part in the bond are locked, not every unit in the molecule.

Section B - Stoichiometry, moles and gas laws (Q14-27)

14. How many molecules are present in 4.4 g of propane, C_3H_8 ? (H = 1, C = 12; Avogadro constant = $6.02 \times 10^{23} \text{ mol}^{-1}$)

- A)** 1.204×10^{23} **B)** 3.01×10^{22} **C)** 6.02×10^{21} **D)** 6.02×10^{22} **E)** 6.02×10^{23}

Answer: D

Given: 4.4 g of propane, C_3H_8 (H = 1, C = 12); Avogadro constant $6.02 \times 10^{23} \text{ mol}^{-1}$

Relationship: amount = mass $\div$ molar mass, then molecules = amount $\times$ Avogadro constant; $M(C_3H_8) = 3(12) + 8(1) = 44 \text{ g/mol}$.

Substitute: $n = 4.4 \text{ g} \div 44 \text{ g/mol}$

Step 1: $n = 0.10 \text{ mol}$ of C_3H_8

Step 2: molecules = $0.10 \text{ mol} \times 6.02 \times 10^{23} \text{ mol}^{-1} = 6.02 \times 10^{22}$

Result: 6.02×10^{22} molecules.

Classic slip: 6.02×10^{23} treats 4.4 g as a whole mole (as if the molar mass were 4.4 g/mol) and overshoots by a factor of ten.

15. How many hydrogen atoms are present in 0.25 mol of water, H_2O ? (Avogadro constant = $6.02 \times 10^{23} \text{ mol}^{-1}$)

- A)** 1.204×10^{24} **B)** 1.505×10^{23} **C)** 3.01×10^{23} **D)** 6.02×10^{23} **E)** 9.03×10^{23}

Answer: C

Given: 0.25 mol of H_2O ; each water molecule carries 2 hydrogen atoms; Avogadro constant $6.02 \times 10^{23} \text{ mol}^{-1}$.

Relationship: moles of H atoms = moles of $H_2O \times 2$, then atoms = moles of H $\times$ Avogadro constant.

Substitute: $n(H) = 0.25 \text{ mol} \times 2$

Step 1: $n(H) = 0.50 \text{ mol}$ of H atoms

Step 2: H atoms = $0.50 \text{ mol} \times 6.02 \times 10^{23} \text{ mol}^{-1} = 3.01 \times 10^{23}$

Result: 3.01×10^{23} hydrogen atoms.

Classic slip: 1.505×10^{23} counts the water molecules and drops the factor of two; 9.03×10^{23} counts all three atoms per molecule instead of the hydrogens alone.

16. What mass of ammonia, NH_3 , contains the same number of molecules as 4.4 g of carbon dioxide, CO_2 ? (C = 12, O = 16, N = 14, H = 1)

- A)** 1.7 g **B)** 3.4 g **C)** 4.4 g **D)** 8.5 g **E)** 17 g

Answer: A

Given: 4.4 g of CO_2 (C = 12, O = 16); equal molecule numbers mean equal moles of NH_3 (N = 14, H = 1).

Relationship: $n(\text{CO}_2) = \text{mass} \div M$, carry the moles across, then mass of $\text{NH}_3 = n \times M$.

Substitute: $M(\text{CO}_2) = 12 + 2(16) = 44 \text{ g/mol}$, so $n = 4.4 \text{ g} \div 44 \text{ g/mol} = 0.10 \text{ mol}$

Step 1: $n(\text{NH}_3) = 0.10 \text{ mol}$ (same molecule count)

Step 2: $M(\text{NH}_3) = 14 + 3(1) = 17 \text{ g/mol}$

Step 3: mass = $0.10 \text{ mol} \times 17 \text{ g/mol} = 1.7 \text{ g}$

Result: 1.7 g of NH_3 .

Classic slip: 4.4 g matches masses instead of molecule counts; equal numbers of molecules need equal moles, and the two molar masses here differ.

17. What volume does 8.8 g of carbon dioxide, CO_2 , occupy at STP? (C = 12, O = 16; molar volume at STP = $22.4 \text{ dm}^3/\text{mol}$)

- A)** 4.48 dm^3 **B)** 8.8 dm^3 **C)** 11.2 dm^3 **D)** 22.4 dm^3 **E)** 44.8 dm^3

Answer: A

Given: 8.8 g of CO_2 (C = 12, O = 16); molar volume at STP = $22.4 \text{ dm}^3/\text{mol}$

Relationship: $n = \text{mass} \div \text{molar mass}$, then $V = n \times \text{molar volume}$.

Substitute: $M(\text{CO}_2) = 12 + 2(16) = 44 \text{ g/mol}$, so $n = 8.8 \text{ g} \div 44 \text{ g/mol} = 0.20 \text{ mol}$

Step 1: $n = 0.20 \text{ mol}$

Step 2: $V = 0.20 \text{ mol} \times 22.4 \text{ dm}^3/\text{mol} = 4.48 \text{ dm}^3$

Result: 4.48 dm^3 at STP.

Classic slip: 11.2 dm^3 assumes 0.5 mol of gas, which would require 22 g of CO_2 , double the stated mass.

18. How many molecules are present in 560 cm^3 of an ideal gas at STP? ($1 \text{ dm}^3 = 1000 \text{ cm}^3$; molar volume at STP = $22.4 \text{ dm}^3/\text{mol}$; Avogadro constant = $6.02 \times 10^{23} \text{ mol}^{-1}$)

- A)** 1.505×10^{22} **B)** 1.505×10^{24} **C)** 1.505×10^{25} **D)** 6.02×10^{21} **E)** 6.02×10^{23}

Answer: A

Given: 560 cm^3 of ideal gas at STP ($1 \text{ dm}^3 = 1000 \text{ cm}^3$); molar volume $22.4 \text{ dm}^3/\text{mol}$; Avogadro constant $6.02 \times 10^{23} \text{ mol}^{-1}$.

Relationship: convert to dm^3 , then $n = V \div \text{molar volume}$, then molecules = $n \times \text{Avogadro constant}$.

Substitute: $V = 560 \text{ cm}^3 \div 1000 = 0.560 \text{ dm}^3$

Step 1: $n = 0.560 \text{ dm}^3 \div 22.4 \text{ dm}^3/\text{mol} = 0.025 \text{ mol}$

Step 2: molecules = $0.025 \text{ mol} \times 6.02 \times 10^{23} \text{ mol}^{-1} = 1.505 \times 10^{22}$

Result: 1.505×10^{22} molecules.

Classic slip: 1.505×10^{25} skips the cm^3 to dm^3 conversion and divides 560 by 22.4 as if the volume were already in cubic decimetres, inflating the count a thousandfold.

19. What is the mass of 4.48 dm^3 of oxygen gas, O_2 , measured at STP? ($\text{O} = 16$; molar volume at STP = $22.4 \text{ dm}^3/\text{mol}$)

- A)** 1.6 g **B)** 3.2 g **C)** 6.4 g **D)** 16 g **E)** 32 g

Answer: C

Given: 4.48 dm^3 of O_2 at STP ($\text{O} = 16$); molar volume $22.4 \text{ dm}^3/\text{mol}$

Relationship: $n = V \div \text{molar volume}$, then $\text{mass} = n \times M$, with $M(\text{O}_2) = 2 \times 16$.

Substitute: $n = 4.48 \text{ dm}^3 \div 22.4 \text{ dm}^3/\text{mol}$

Step 1: $n = 0.20 \text{ mol}$

Step 2: $M(\text{O}_2) = 2 \times 16 = 32 \text{ g/mol}$

Step 3: $\text{mass} = 0.20 \text{ mol} \times 32 \text{ g/mol} = 6.4 \text{ g}$

Result: 6.4 g.

Classic slip: 3.2 g uses the atomic mass 16 g/mol and forgets that oxygen gas is diatomic.

20. 10.0 g of calcium carbonate ($M = 100 \text{ g/mol}$) is treated with 100 cm^3 of 1.0 mol/dm^3 hydrochloric acid: $\text{CaCO}_3 + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2\text{O} + \text{CO}_2$. What volume of CO_2 is released, measured at STP? ($1 \text{ dm}^3 = 1000 \text{ cm}^3$; molar volume at STP = $22.4 \text{ dm}^3/\text{mol}$)

- A)** 0.56 dm^3 **B)** 1.12 dm^3 **C)** 2.24 dm^3 **D)** 11.2 dm^3 **E)** 22.4 dm^3

Answer: B

Given: 10.0 g of CaCO_3 ($M = 100 \text{ g/mol}$); $100 \text{ cm}^3 = 0.100 \text{ dm}^3$ of 1.0 mol/dm^3 HCl; $\text{CaCO}_3 + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2\text{O} + \text{CO}_2$; molar volume $22.4 \text{ dm}^3/\text{mol}$.

Relationship: convert both reactants to moles, apply the 1:2 stoichiometry to find the limiting reactant, then 1 mol CO_2 per mol of carbonate actually reacted.

Substitute: $n(\text{CaCO}_3) = 10.0 \text{ g} \div 100 \text{ g/mol} = 0.10 \text{ mol}$; $n(\text{HCl}) = 0.100 \text{ dm}^3 \times 1.0 \text{ mol/dm}^3 = 0.10 \text{ mol}$

Step 1: reacting all the carbonate would need $2 \times 0.10 = 0.20 \text{ mol}$ of HCl.

Step 2: only 0.10 mol of HCl is available, so the acid runs out first.

Step 3: from $2\text{HCl} : 1\text{CO}_2$, $n(\text{CO}_2) = 0.10 \text{ mol} \div 2 = 0.050 \text{ mol}$

Step 4: $V = 0.050 \text{ mol} \times 22.4 \text{ dm}^3/\text{mol} = 1.12 \text{ dm}^3$

Result: 1.12 dm^3 of CO_2 at STP.

Classic slip: 2.24 dm^3 treats the carbonate as the limiting reactant and ignores that every mole of CO_2 consumes two moles of HCl.

21. In theory, 9.2 g of ethanol ($M = 46 \text{ g/mol}$) dehydrates completely to ethene ($M = 28 \text{ g/mol}$): $\text{C}_2\text{H}_5\text{OH} \rightarrow \text{C}_2\text{H}_4 + \text{H}_2\text{O}$. The actual mass of ethene collected is 4.2 g. What is the percentage yield?

- A)** 25% **B)** 56% **C)** 75% **D)** 100% **E)** 133%

Answer: C

Given: 9.2 g of ethanol ($M = 46 \text{ g/mol}$); $\text{C}_2\text{H}_5\text{OH} \rightarrow \text{C}_2\text{H}_4 + \text{H}_2\text{O}$ in a 1:1 ratio; actual ethene collected 4.2 g ($M = 28 \text{ g/mol}$).

Relationship: percentage yield = (actual mass $\div$ theoretical mass) $\times$ 100, with the theoretical mass fixed by the 1:1 mole ratio.

Substitute: $n(\text{ethanol}) = 9.2 \text{ g} \div 46 \text{ g/mol} = 0.20 \text{ mol}$, so $n(\text{theoretical ethene}) = 0.20 \text{ mol}$

Step 1: theoretical mass = $0.20 \text{ mol} \times 28 \text{ g/mol} = 5.6 \text{ g}$

Step 2: percentage yield = $(4.2 \text{ g} \div 5.6 \text{ g}) \times 100$

Step 3: $4.2 \div 5.6 = 0.75$, so the yield is 75%

Result: 75%.

Classic slip: 25% reports the 1.4 g shortfall as a share of the theoretical mass instead of the fraction actually obtained, and 133% inverts the ratio.

22. A diver exhales an air bubble of volume 2.0 dm^3 at a depth where the pressure is 400 kPa. The temperature stays constant as the bubble rises to the surface, where the pressure is 100 kPa. What is the volume of the bubble at the surface?

- A)** 0.5 dm^3 **B)** 1.0 dm^3 **C)** 2.0 dm^3 **D)** 4.0 dm^3 **E)** 8.0 dm^3

Answer: E

Given: bubble volume 2.0 dm^3 at depth pressure 400 kPa; surface pressure 100 kPa; temperature constant.

Relationship: Boyle's law at constant temperature: $P_1V_1 = P_2V_2$, so $V_2 = P_1V_1 \div P_2$.

Substitute: $V_2 = (400 \text{ kPa} \times 2.0 \text{ dm}^3) \div 100 \text{ kPa}$

Step 1: $P_1V_1 = 400 \times 2.0 = 800 \text{ kPa} \cdot \text{dm}^3$

Step 2: $V_2 = 800 \div 100 = 8.0 \text{ dm}^3$

Result: 8.0 dm^3 at the surface.

Classic slip: 0.5 dm^3 inverts the pressure ratio and wrongly predicts the bubble shrinking as it rises into lower pressure.

23. A rigid sealed canister holds a fixed amount of gas at 27°C and 100 kPa. It is heated until the gas reaches 327°C . What is the new pressure inside the canister?

- A)** 50 kPa **B)** 100 kPa **C)** 109 kPa **D)** 200 kPa **E)** 300 kPa

Answer: D

Given: rigid canister (fixed volume and moles) at 27°C and 100 kPa, heated to 327°C .

Relationship: pressure-temperature law $P_1 \div T_1 = P_2 \div T_2$, so $P_2 = P_1 \times T_2 \div T_1$, with T in kelvin ($T = ^\circ\text{C} + 273$).

Substitute: $P_2 = 100 \text{ kPa} \times (T_2 \div T_1)$

Step 1: $T_1 = 27 + 273 = 300 \text{ K}$; $T_2 = 327 + 273 = 600 \text{ K}$

Step 2: $T_2 \div T_1 = 600 \div 300 = 2$

Step 3: $P_2 = 100 \text{ kPa} \times 2 = 200 \text{ kPa}$

Result: 200 kPa.

Classic slip: 109 kPa builds the ratio from the Celsius readings ($327 \div 300$), which are not absolute temperatures.

24. A gas occupies 9.0 dm^3 at 27°C and 100 kPa . What volume does it occupy at 127°C and 200 kPa ?

- A) 4.5 dm^3 B) 6.0 dm^3 C) 12 dm^3 D) 13.5 dm^3 E) 24 dm^3

Answer: B

Given: 9.0 dm^3 at 27°C and 100 kPa , taken to 127°C and 200 kPa .

Relationship: combined gas law $V_2 = V_1 \times (T_2 \div T_1) \times (P_1 \div P_2)$, temperatures in kelvin.

Substitute: $T_1 = 300 \text{ K}$ and $T_2 = 400 \text{ K}$, so $V_2 = 9.0 \text{ dm}^3 \times (400 \div 300) \times (100 \div 200)$

Step 1: temperature factor = $400 \div 300 \approx 1.33$

Step 2: pressure factor = $100 \div 200 = 0.50$

Step 3: $V_2 = 9.0 \times 1.33 \times 0.50 = 6.0 \text{ dm}^3$

Result: 6.0 dm^3 (warming expands the gas, but the doubled pressure compresses it more).

Classic slip: 12 dm^3 keeps only the warming factor and ignores the pressure doubling; 13.5 dm^3 inverts both ratios.

25. A sealed vessel contains 0.20 mol of nitrogen gas and 0.30 mol of oxygen gas at constant temperature, and the total pressure is 150 kPa . What is the partial pressure of the oxygen?

- A) 45 kPa B) 60 kPa C) 75 kPa D) 90 kPa E) 150 kPa

Answer: D

Given: 0.20 mol of N_2 plus 0.30 mol of O_2 in one vessel; total pressure 150 kPa ; constant temperature.

Relationship: Dalton's law: partial pressure = mole fraction $\times$ total pressure, with $x(\text{O}_2) = n(\text{O}_2) \div n(\text{total})$.

Substitute: $n(\text{total}) = 0.20 + 0.30 = 0.50 \text{ mol}$

Step 1: $x(\text{O}_2) = 0.30 \text{ mol} \div 0.50 \text{ mol} = 0.60$

Step 2: $p(\text{O}_2) = 0.60 \times 150 \text{ kPa} = 90 \text{ kPa}$

Result: 90 kPa .

Classic slip: 45 kPa multiplies the raw 0.30 mol by the total pressure without converting to a mole fraction first, and 60 kPa is the nitrogen share.

26. 100 cm^3 of 0.80 mol/dm^3 hydrochloric acid is mixed with 400 cm^3 of 0.20 mol/dm^3 hydrochloric acid. Assuming the volumes are additive, what is the concentration of the resulting solution?

- A) 0.16 mol/dm^3 B) 0.20 mol/dm^3 C) 0.32 mol/dm^3 D) 0.50 mol/dm^3 E) 0.80 mol/dm^3

Answer: C

Given: $100 \text{ cm}^3 = 0.100 \text{ dm}^3$ of 0.80 mol/dm^3 HCl mixed with $400 \text{ cm}^3 = 0.400 \text{ dm}^3$ of 0.20 mol/dm^3 HCl; volumes additive.

Relationship: total moles = $\Sigma(\text{volume} \times \text{concentration})$ over both stocks; final concentration = total moles $\div$ total volume.

Substitute: $n_1 = 0.100 \text{ dm}^3 \times 0.80 \text{ mol/dm}^3$; $n_2 = 0.400 \text{ dm}^3 \times 0.20 \text{ mol/dm}^3$

Step 1: $n_1 = 0.080 \text{ mol}$

Step 2: $n_2 = 0.080 \text{ mol}$

Step 3: $n(\text{total}) = 0.080 + 0.080 = 0.160 \text{ mol}$; $V(\text{total}) = 0.100 + 0.400 = 0.500 \text{ dm}^3$

Step 4: $c = 0.160 \text{ mol} \div 0.500 \text{ dm}^3 = 0.32 \text{ mol/dm}^3$

Result: 0.32 mol/dm^3 .

Classic slip: 0.50 mol/dm^3 takes a plain average of the two concentrations and ignores that four times more of the dilute solution entered the mixture.

27. A solution is prepared by diluting 25 cm³ of a 2.0 mol/dm³ solution to a concentration of 0.10 mol/dm³. What is the final volume of the diluted solution?

- A)** 25 cm³ **B)** 250 cm³ **C)** 450 cm³ **D)** 500 cm³ **E)** 1250 cm³

Answer: D

Given: 25 cm³ = 0.025 dm³ of 2.0 mol/dm³ solution, diluted to 0.10 mol/dm³

Relationship: dilution conserves moles of solute: $c_1V_1 = c_2V_2$, so $V_2 = c_1V_1 \div c_2$.

Substitute: moles of solute = 0.025 dm³ × 2.0 mol/dm³ = 0.050 mol

Step 1: $V_2 = 0.050 \text{ mol} \div 0.10 \text{ mol/dm}^3 = 0.50 \text{ dm}^3$

Step 2: 0.50 dm³ × 1000 = 500 cm³

Result: 500 cm³ of diluted solution.

Classic slip: subtracting the original 25 cm³ answers the water-added question instead; the final volume is the whole diluted solution, and the 25 cm³ of stock already sits inside it.

Section C - Solutions, acids and bases (Q28-40)

28. What is the pH of a 2.0×10^{-4} mol/dm³ solution of hydrochloric acid, a strong acid that is fully ionised in water?

- A)** 2.7 **B)** 3.3 **C)** 3.7 **D)** 4.0 **E)** 10.3

Answer: C

Given: 2.0×10^{-4} mol/dm³ HCl, fully ionised, so $[H^+] = 2.0 \times 10^{-4}$ mol/dm³

Relationship: $\text{pH} = -\log[H^+]$; for a coefficient times a power of ten, $-\log(a \times 10^{-n}) = n - \log a$.

Substitute: $\text{pH} = 4 - \log 2.0$

Step 1: $\log 2.0 = 0.30$

Step 2: $\text{pH} = 4 - 0.30 = 3.70$

Result: pH 3.70.

Classic slip: 4.0 drops the factor of 2 (its 0.30 log contribution) and answers as if $[H^+]$ were exactly 1×10^{-4} mol/dm³.

29. What is the pOH of a 5.0×10^{-2} mol/dm³ solution of nitric acid, a strong acid that is fully ionised in water? ($K_w = 1.0 \times 10^{-14}$ at 25 °C)

- A)** 1.3 **B)** 2.7 **C)** 12.3 **D)** 12.7 **E)** 13.7

Answer: D

Given: 5.0×10^{-2} mol/dm³ HNO₃, fully ionised, so $[H^+] = 5.0 \times 10^{-2}$ mol/dm³; $K_w = 1.0 \times 10^{-14}$ at 25 °C

Relationship: $\text{pH} = -\log[H^+]$, then $\text{pOH} = 14.00 - \text{pH}$ (from $K_w = [H^+][OH^-]$).

Substitute: $\text{pH} = -\log(5.0 \times 10^{-2}) = 2 - \log 5.0$

Step 1: $\log 5.0 = 0.70$, so $\text{pH} = 2 - 0.70 = 1.30$

Step 2: $\text{pOH} = 14.00 - 1.30 = 12.70$

Result: pOH 12.70.

Classic slip: 1.30 is the pH, not the pOH; the crossing to the hydroxide scale must run through K_w .

30. Strontium hydroxide, $\text{Sr}(\text{OH})_2$, is a strong base that is fully dissociated in water. What is the pH of a $5.0 \times 10^{-4} \text{ mol/dm}^3$ solution of $\text{Sr}(\text{OH})_2$? ($K_w = 1.0 \times 10^{-14}$ at 25°C)

A) 3.0 **B)** 10.7 **C)** 11.0 **D)** 11.7 **E)** 12.0

Answer: C

Given: $5.0 \times 10^{-4} \text{ mol/dm}^3$ $\text{Sr}(\text{OH})_2$, fully dissociated; each formula unit releases 2 OH^- ; $K_w = 1.0 \times 10^{-14}$ at 25°C .

Relationship: $[\text{OH}^-] = 2 \times \text{concentration}$, then $\text{pOH} = -\log[\text{OH}^-]$, then $\text{pH} = 14.00 - \text{pOH}$.

Substitute: $[\text{OH}^-] = 2 \times 5.0 \times 10^{-4} \text{ mol/dm}^3$

Step 1: $[\text{OH}^-] = 1.0 \times 10^{-3} \text{ mol/dm}^3$

Step 2: $\text{pOH} = -\log(1.0 \times 10^{-3}) = 3.00$

Step 3: $\text{pH} = 14.00 - 3.00 = 11.00$

Result: pH 11.0.

Classic slip: 10.7 forgets the factor of two and carries pOH 3.30 from a single OH^- per formula unit; 3.0 quotes the pOH as if it were the pH of a base.

31. Four solutions are prepared at 25°C ($K_w = 1.0 \times 10^{-14}$) from fully ionised solutes: P is $1.0 \times 10^{-3} \text{ mol/dm}^3$ HCl, Q is $5.0 \times 10^{-3} \text{ mol/dm}^3$ HCl, R is $1.0 \times 10^{-3} \text{ mol/dm}^3$ NaOH and S is $1.0 \times 10^{-3} \text{ mol/dm}^3$ $\text{Ba}(\text{OH})_2$. Which option places the four solutions in order of increasing pH?

A) Q, P, S, R **B)** P, Q, R, S **C)** S, R, Q, P **D)** P, Q, S, R **E)** Q, P, R, S

Answer: E

Given: P is $1.0 \times 10^{-3} \text{ mol/dm}^3$ HCl; Q is $5.0 \times 10^{-3} \text{ mol/dm}^3$ HCl; R is $1.0 \times 10^{-3} \text{ mol/dm}^3$ NaOH; S is $1.0 \times 10^{-3} \text{ mol/dm}^3$ $\text{Ba}(\text{OH})_2$; $K_w = 1.0 \times 10^{-14}$; all solutes fully ionised.

Relationship: acids: $\text{pH} = -\log[\text{H}^+]$; bases: $\text{pOH} = -\log[\text{OH}^-]$ and $\text{pH} = 14 - \text{pOH}$; $\text{Ba}(\text{OH})_2$ releases 2 OH^- per formula unit.

Step 1: P: $[\text{H}^+] = 1.0 \times 10^{-3} \text{ mol/dm}^3$, pH 3.00

Step 2: Q: $[\text{H}^+] = 5.0 \times 10^{-3} \text{ mol/dm}^3$, pH = $3 - 0.70 = 2.30$

Step 3: R: $[\text{OH}^-] = 1.0 \times 10^{-3} \text{ mol/dm}^3$, pOH 3.00, pH 11.00

Step 4: S: $[\text{OH}^-] = 2 \times 10^{-3} \text{ mol/dm}^3$, pOH = $3 - 0.30 = 2.70$, pH 11.30

Result: increasing pH runs Q (2.3), then P (3.0), then R (11.0), then S (11.3).

Classic slip: placing P ahead of Q orders by concentration alone and forgets that the logarithm makes the five-times-more-concentrated acid the more acidic one.

32. Pure water is heated from 25°C to 60°C . As the temperature rises, K_w increases, and the pH of pure water at 60°C is measured to be 6.5. Which of the following statements is correct?

- A)** The water is basic, because heating pushes the hydroxide concentration above the hydrogen ion concentration.
B) The water is acidic, because a pH below 7 always means more H_3O^+ than OH^- .
C) The water is neutral, because K_w keeps the value 1.0×10^{-14} at every temperature.
D) The water is acidic, because heating destroys hydroxide ions faster than hydrogen ions.
E) The water is neutral, because $[\text{H}^+]$ and $[\text{OH}^-]$ stay equal while both become larger than at 25°C .

Answer: E

Pure water always produces H_3O^+ and OH^- in equal amounts, so it remains neutral at every temperature; neutrality means equal concentrations of the two ions, not a pH of exactly 7, which is a 25°C coincidence tied to $K_w = 1.0 \times 10^{-14}$. Heating raises K_w , so both concentrations climb above 10^{-7} mol/dm^3 and the pH falls to 6.5 without any excess of H_3O^+ over OH^- .

33. At 10 °C the ionic product of water has fallen to $K_w = 3.0 \times 10^{-15} \text{ mol}^2 \text{ dm}^{-6}$. What is the hydroxide ion concentration, $[\text{OH}^-]$, in pure water at 10 °C?

- A)** $1.0 \times 10^{-7} \text{ mol/dm}^3$
- B)** $1.7 \times 10^{-8} \text{ mol/dm}^3$
- C)** $3.0 \times 10^{-15} \text{ mol/dm}^3$
- D)** $5.5 \times 10^{-8} \text{ mol/dm}^3$
- E)** $9.0 \times 10^{-8} \text{ mol/dm}^3$

Answer: D

Given: pure water at 10 °C with $K_w = 3.0 \times 10^{-15} \text{ mol}^2 \text{ dm}^{-6}$; in pure water $[\text{H}^+] = [\text{OH}^-]$.

Relationship: $K_w = [\text{H}^+][\text{OH}^-] = [\text{OH}^-]^2$, so $[\text{OH}^-] = \sqrt{K_w}$

Substitute: $[\text{OH}^-] = \sqrt{(3.0 \times 10^{-15}) \text{ mol/dm}^3}$

Step 1: rewrite 3.0×10^{-15} as 30×10^{-16} .

Step 2: $\sqrt{30} \approx 5.5$ and $\sqrt{(10^{-16})} = 10^{-8}$, so $[\text{OH}^-] \approx 5.5 \times 10^{-8} \text{ mol/dm}^3$

Result: $[\text{OH}^-] \approx 5.5 \times 10^{-8} \text{ mol/dm}^3$.

Classic slip: 1.0×10^{-7} recycles the 25 °C concentration although K_w has dropped; 3.0×10^{-15} mistakes the ionic product itself for a concentration.

34. 10 cm³ of 0.10 mol/dm³ hydrochloric acid, fully ionised, is diluted with water to a final volume of 1.0 dm³. By how much does the pH of the solution change?

- A)** It stays the same, because the amount of acid is unchanged.
- B)** It reaches pH 7, because a hundredfold dilution turns any strong acid neutral.
- C)** It falls by 2 units.
- D)** It rises by 1 unit.
- E)** It rises by 2 units.

Answer: E

Given: 10 cm³ = 0.010 dm³ of 0.10 mol/dm³ HCl, fully ionised, diluted to a final volume of 1.0 dm³.

Relationship: the mole count of H⁺ is fixed by the stock; new $[\text{H}^+] = \text{moles} \div \text{new volume}$; $\text{pH} = -\log[\text{H}^+]$.

Substitute: moles of H⁺ = 0.010 dm³ × 0.10 mol/dm³ = $1.0 \times 10^{-3} \text{ mol}$

Step 1: before dilution, $[\text{H}^+] = 0.10 \text{ mol/dm}^3$, pH 1.0

Step 2: after dilution, $[\text{H}^+] = 1.0 \times 10^{-3} \text{ mol} \div 1.0 \text{ dm}^3 = 1.0 \times 10^{-3} \text{ mol/dm}^3$

Step 3: $\text{pH} = -\log(1.0 \times 10^{-3}) = 3.0$

Result: the pH rises by $3.0 - 1.0 = 2$ units

Classic slip: expecting pH 7 treats dilution as neutralisation; a hundredfold dilution only cuts $[\text{H}^+]$ a hundredfold, and 10^{-3} mol/dm^3 of strong acid still has pH 3.

35. What volume of water must be added to 20 cm³ of 1.0 × 10⁻² mol/dm³ hydrochloric acid, fully ionised, to obtain a solution of pH 4.0 at 25 °C?

- A)** 180 cm³ **B)** 198 cm³ **C)** 200 cm³ **D)** 1980 cm³ **E)** 2000 cm³

Answer: D

Given: 20 cm³ = 0.020 dm³ of 1.0 × 10⁻² mol/dm³ HCl; target pH 4.0 means target [H⁺] = 1.0 × 10⁻⁴ mol/dm³.

Relationship: moles of H⁺ are conserved, so final volume = moles ÷ target concentration; water added = final volume – stock volume.

Substitute: moles of H⁺ = 0.020 dm³ × 1.0 × 10⁻² mol/dm³ = 2.0 × 10⁻⁴ mol

Step 1: final volume = 2.0 × 10⁻⁴ mol ÷ 1.0 × 10⁻⁴ mol/dm³ = 2.0 dm³ = 2000 cm³

Step 2: water to add = 2000 – 20 = 1980 cm³

Result: 1980 cm³ of water.

Classic slip: 2000 cm³ quotes the total final volume and forgets that the 20 cm³ of stock already occupies part of it.

36. Which of the following mixtures, made up with pure water as solvent, can act as a buffer solution?

- A)** 100 cm³ of 0.10 mol/dm³ ethanoic acid mixed with 100 cm³ of 0.10 mol/dm³ sodium hydroxide.
B) 100 cm³ of 0.10 mol/dm³ ethanoic acid mixed with 50 cm³ of 0.10 mol/dm³ sodium hydroxide.
C) 100 cm³ of 0.10 mol/dm³ hydrochloric acid mixed with 50 cm³ of 0.10 mol/dm³ sodium hydroxide.
D) A 0.10 mol/dm³ solution of ethanoic acid alone in water.
E) 100 cm³ of 0.10 mol/dm³ hydrochloric acid mixed with 100 cm³ of 0.10 mol/dm³ sodium chloride.

Answer: B

Given: 100 cm³ = 0.100 dm³ of 0.10 mol/dm³ ethanoic acid (weak) mixed with 50 cm³ = 0.050 dm³ of 0.10 mol/dm³ NaOH (strong).

Relationship: a buffer needs comparable amounts of a weak acid and its conjugate base; moles = volume × concentration, and OH⁻ neutralises the acid 1:1.

Substitute: n(acid) = 0.100 dm³ × 0.10 mol/dm³ = 0.010 mol; n(OH⁻) = 0.050 dm³ × 0.10 mol/dm³ = 0.0050 mol

Step 1: CH₃COOH + OH⁻ → CH₃COO⁻ + H₂O consumes 0.0050 mol of the acid.

Step 2: acid remaining = 0.010 – 0.0050 = 0.0050 mol; ethanoate formed = 0.0050 mol.

Result: 0.0050 mol of CH₃COOH and 0.0050 mol of CH₃COO⁻ now sit side by side, the half-neutralised weak acid mixture that buffers.

Classic slip: full neutralisation (equal volumes of the acid and the alkali) consumes every mole of acid, leaving only sodium ethanoate with no reserve to absorb added alkali; a weak acid alone carries essentially no conjugate base.

37. A buffer solution contains comparable amounts of ethanoic acid and sodium ethanoate and has pH 4.8. An equal volume of pure water is added to it. Which of the following statements is correct?
- A) The pH rises by about one unit because both concentrations fall by a factor of two.
 - B) The pH falls because the acid becomes effectively stronger on dilution.
 - C) The buffer is destroyed, because dilution pushes the dissociation of the acid completely towards the ions.
 - D) The pH and the buffer capacity are both unchanged, because a buffer resists any change made to it.
 - E) The pH stays close to 4.8 because the ratio of ethanoate to ethanoic acid is unchanged, but the buffer capacity falls.

Answer: E

The pH of an ethanoic acid and ethanoate buffer is governed by the ratio $[\text{CH}_3\text{COO}^-] \div [\text{CH}_3\text{COOH}]$. Adding an equal volume of water divides both concentrations by the same factor, so the ratio, and hence the pH near 4.8, stays essentially unchanged, while the absolute amounts of acid and base available to absorb any added acid or alkali shrink. Dilution costs a buffer its capacity, not its pH, so the claims that the buffer is destroyed or that the pH jumps by about a unit misread which property the ratio controls.

38. Which of the following statements about the behaviour of salts dissolved in water is correct?
- A) Ammonium chloride gives a basic solution, because the ammonium ion accepts a proton from water.
 - B) Sodium ethanoate gives a basic solution, because the ethanoate ion accepts a proton from water and releases OH^- .
 - C) Sodium ethanoate gives an acidic solution, because the sodium ion donates protons to water.
 - D) Sodium chloride gives a basic solution, because the chloride ion accepts protons from water.
 - E) Ammonium nitrate gives a basic solution, because the nitrate ion accepts protons from water.

Answer: B

The ethanoate ion is the conjugate base of a weak acid, so it accepts a proton from water, $\text{CH}_3\text{COO}^- + \text{H}_2\text{O} \rightleftharpoons \text{CH}_3\text{COOH} + \text{OH}^-$, releasing hydroxide and making sodium ethanoate basic. The ammonium claim reverses the chemistry, because NH_4^+ donates a proton to water and gives an acidic solution, and chloride and nitrate, as conjugate bases of strong acids, are far too weak to hydrolyse water.

39. A drip bag holds 500 cm^3 of glucose solution containing 9.0 g of glucose ($M = 180 \text{ g/mol}$); the label reads 1.8 % w/v, meaning 1.8 g of glucose per 100 cm^3 of solution. What is the molarity of the glucose?
- A) 0.010 mol/dm^3 B) 0.050 mol/dm^3 C) 0.100 mol/dm^3 D) 0.180 mol/dm^3 E) 1.80 mol/dm^3

Answer: C

Given: 9.0 g of glucose ($M = 180 \text{ g/mol}$) in $500 \text{ cm}^3 = 0.500 \text{ dm}^3$ of solution; label 1.8 % w/v means 1.8 g per 100 cm^3 .

Relationship: $c = n \div V$, with $n = \text{mass} \div \text{molar mass}$.

Substitute: $n = 9.0 \text{ g} \div 180 \text{ g/mol} = 0.050 \text{ mol}$

Step 1: $c = 0.050 \text{ mol} \div 0.500 \text{ dm}^3 = 0.10 \text{ mol/dm}^3$

Step 2: label cross-check: 1.8 g per 100 cm^3 scales to 9.0 g per 500 cm^3 , consistent with the stem.

Result: 0.10 mol/dm^3 .

Classic slip: 0.050 mol/dm^3 reports the mole count itself as a concentration and skips the division by the bag's volume.

40. At 25 °C the solubility of calcium hydroxide in water is 1.85 g per dm³ of saturated solution, and the dissolved salt is fully dissociated. ($M(\text{Ca}(\text{OH})_2) = 74 \text{ g/mol}$; $K_w = 1.0 \times 10^{-14}$ at 25 °C; for dilute aqueous solutions 1 ppm is 1 mg per dm³.) Which of the following statements is correct?

- A)** The solubility is 0.025 mol/dm³ and the pH of the saturated solution is 12.70.
- B)** The solubility is 0.050 mol/dm³ and the pH of the saturated solution is 13.30.
- C)** The solubility is 0.050 mol/dm³ and the pH of the saturated solution is 12.70.
- D)** The solubility is 0.025 mol/dm³ and the calcium ion concentration is 1850 ppm.
- E)** The solubility is 0.025 mol/dm³ and the pH of the saturated solution is 11.30.

Answer: A

Given: solubility 1.85 g of $\text{Ca}(\text{OH})_2$ per dm³ ($M = 74 \text{ g/mol}$); fully dissociated; $K_w = 1.0 \times 10^{-14}$ at 25 °C; 1 ppm = 1 mg per dm³.

Relationship: molar solubility = mass $\div$ M ; $\text{Ca}(\text{OH})_2 \rightarrow \text{Ca}^{2+} + 2\text{OH}^-$ so $[\text{OH}^-] = 2 \times \text{solubility}$; $\text{pOH} = -\log[\text{OH}^-]$; $\text{pH} = 14 - \text{pOH}$.

Substitute: $s = 1.85 \text{ g} \div 74 \text{ g/mol} = 0.025 \text{ mol/dm}^3$

Step 1: $[\text{OH}^-] = 2 \times 0.025 = 0.050 \text{ mol/dm}^3$

Step 2: $\text{pOH} = -\log(5.0 \times 10^{-2}) = 2 - 0.70 = 1.30$

Step 3: $\text{pH} = 14.00 - 1.30 = 12.70$

Result: solubility 0.025 mol/dm³ with pH 12.70.

Classic slip: 0.050 mol/dm³ reports the hydroxide concentration as the salt's solubility; the 1850 ppm claim also fails, since Ca^{2+} amounts to $0.025 \text{ mol} \times 40 \text{ g/mol} = 1.0 \text{ g/dm}^3 = 1000 \text{ ppm}$.

Section D - Redox and oxidation numbers (Q41-49)

41. What is the oxidation number of the carbon atom in methanal, HCHO ?

- A)** -2 **B)** 0 **C)** +1 **D)** +2 **E)** +4

Answer: B

Given: methanal, HCHO , a neutral molecule; H counts +1 and O counts -2 in compounds.

Relationship: oxidation numbers sum to zero over a neutral molecule: $n(\text{C}) + 2 \times n(\text{H}) + n(\text{O}) = 0$.

Substitute: $n(\text{C}) + 2(+1) + (-2) = 0$

Step 1: $n(\text{C}) + 2 - 2 = 0$

Step 2: $n(\text{C}) + 0 = 0$

Result: the carbon sits at oxidation number 0.

Classic slip: +2 appears when the two +1 hydrogens are left out of the sum, leaving the carbon to balance the oxygen alone; +2 actually belongs to the carbon in CO , and +4 to CO_2 .

42. In the thiosulfate ion, $\text{S}_2\text{O}_3^{2-}$, the structure has a central sulfur atom bonded to three oxygen atoms and to one terminal sulfur atom, as if one oxygen of a sulfate unit had been replaced by sulfur. Which of the following statements about oxidation numbers in this ion is correct?

- A)** The average oxidation number of sulfur is +2, with the central sulfur at +5 and the terminal sulfur at -1.
- B)** The average oxidation number of sulfur is +4, because each sulfur behaves as in the sulfate ion.
- C)** Both sulfur atoms have oxidation number +2, because the charge of the ion is -2.
- D)** The terminal sulfur has oxidation number -2 and the central sulfur +4, exactly as in sulfate.
- E)** Oxidation numbers cannot be assigned, because the bond between identical atoms makes all the numbers undefined.

Answer: A

Given: thiosulfate ion $\text{S}_2\text{O}_3^{2-}$; central S bonded to three O atoms and to one terminal S; O is -2; the oxidation numbers must sum to the ion charge, -2.

Relationship: $n(\text{central}) + n(\text{terminal}) + 3(-2) = -2$; a bond between identical atoms counts 0 because the bonding electrons are shared equally.

Substitute: $n(\text{central}) + n(\text{terminal}) - 6 = -2$

Step 1: $n(\text{central}) + n(\text{terminal}) = +4$

Step 2: average per sulfur = $+4 \div 2 = +2$

Step 3: split by structure: with the S-S bond ignored, the central sulfur facing the three oxygens takes +5, and the terminal sulfur takes -1.

Step 4: check: $+5 + (-1) + 3(-2) = +5 - 1 - 6 = -2 \checkmark$

Result: average +2, central +5, terminal -1.

Classic slip: +4 for the average copies the sulfate value; swapping a sulfur for an oxygen lowers the total the two sulfurs must carry, and an identical-atom bond is ignored in the count, not undefined.

43. Which of the following statements about oxidation numbers in oxoanions is correct?

- A)** Sulfur is +7 in SO_4^{2-} and nitrogen is +4 in NO_2^- .
- B)** Sulfur is +6 in SO_4^{2-} and nitrogen is +5 in NO_2^- .
- C)** Sulfur is +4 in SO_4^{2-} and nitrogen is +3 in NO_2^- .
- D)** Sulfur is 0 in SO_4^{2-} because the oxygens carry the whole charge, and nitrogen is +3 in NO_2^- .
- E)** Sulfur is +6 in SO_4^{2-} and nitrogen is +3 in NO_2^- .

Answer: E

Given: sulfate SO_4^{2-} and nitrite NO_2^- ; O is -2; each sum must equal that ion's charge.

Relationship: $n(\text{X}) + (\text{number of O}) \times (-2) = \text{ion charge}$; solve for $n(\text{X})$.

Step 1: sulfate: $n(\text{S}) + 4(-2) = -2$, so $n(\text{S}) = -2 + 8 = +6$

Step 2: nitrite: $n(\text{N}) + 2(-2) = -1$, so $n(\text{N}) = -1 + 4 = +3$

Result: sulfur is +6 in SO_4^{2-} and nitrogen is +3 in NO_2^- .

Classic slip: +4 for sulfur belongs to sulfite, SO_3^{2-} , one oxygen fewer, and +5 for nitrogen belongs to nitrate, NO_3^- , one oxygen more; swapping the oxoanions swaps the numbers.

44. Ethanol, $\text{CH}_3\text{CH}_2\text{OH}$, is oxidised to ethanal, CH_3CHO . Which of the following statements is correct?
- A) The carbon bearing the OH group goes from oxidation number -1 to $+1$, so ethanol is oxidised.
 - B) Both carbon atoms rise from -3 to $+1$, so ethanol is oxidised.
 - C) The oxidation number of oxygen rises from -2 to 0 , so oxygen is oxidised.
 - D) The average oxidation number of carbon is unchanged, so no redox change occurs.
 - E) The carbon bearing the OH group goes from oxidation number $+1$ to -1 , so ethanol is reduced.

Answer: A

Given: ethanol $\text{CH}_3\text{CH}_2\text{OH}$ oxidised to ethanal CH_3CHO ; bond contributions to a carbon's oxidation number: C-H -1 , C-O $+1$, C=O $+2$, C-C 0 .

Relationship: the carbon's oxidation number = sum of the contributions of its bonds.

Step 1: ethanol, the CH_2OH carbon: $2(\text{C-H}) + 1(\text{C-O}) + 1(\text{C-C}) = 2(-1) + (+1) + 0 = -1$

Step 2: ethanal, the CHO carbon: $1(\text{C-H}) + 1(\text{C=O}) + 1(\text{C-C}) = (-1) + (+2) + 0 = +1$

Step 3: change = $+1 - (-1) = +2$, a loss of 2 electrons

Result: the carbon rises from -1 to $+1$, so ethanol is oxidised.

Classic slip: reading the change backwards (from $+1$ to -1) describes reduction; the CH_3 carbon stays at -3 and is a spectator, so any claim that both carbons rise is a miscount.

45. In aqueous solution, $2\text{FeCl}_3 + \text{SnCl}_2 \rightarrow 2\text{FeCl}_2 + \text{SnCl}_4$. Which species acts as the oxidising agent?
- A) FeCl_2 , because iron ends at oxidation number $+2$.
 - B) FeCl_3 , because Fe^{3+} is reduced to Fe^{2+} .
 - C) Cl^- , because it keeps oxidation number -1 in every compound.
 - D) SnCl_2 , because tin is reduced from $+2$ to $+4$.
 - E) SnCl_4 , because tin is oxidised from $+2$ to $+4$.

Answer: B

Given: $2\text{FeCl}_3 + \text{SnCl}_2 \rightarrow 2\text{FeCl}_2 + \text{SnCl}_4$; Cl stays -1 throughout.

Relationship: the species that is reduced (gains electrons, oxidation number falls) is by definition the oxidising agent.

Step 1: Sn goes from $+2$ in SnCl_2 to $+4$ in SnCl_4 , a rise of 2, so tin is oxidised and SnCl_2 is the reducing agent.

Step 2: Fe goes from $+3$ in FeCl_3 to $+2$ in FeCl_2 , a fall of 1 per iron, so Fe^{3+} gains electrons and is reduced.

Result: FeCl_3 is reduced, so FeCl_3 is the oxidising agent.

Classic slip: naming the oxidation product as the agent reverses the rule; the species being reduced is the one doing the oxidising.

46. In the laboratory preparation of chlorine, MnO_2 reacts with concentrated hydrochloric acid: $\text{MnO}_2 + 4\text{HCl} \rightarrow \text{MnCl}_2 + \text{Cl}_2 + 2\text{H}_2\text{O}$. Which of the following statements is correct?

- A)** HCl is the oxidising agent, because it supplies the hydrogen ions.
- B)** MnO_2 is the reducing agent, because manganese goes from +4 to +2.
- C)** Chlorine gas is the reducing agent, because it is the oxidation product.
- D)** No oxidation occurs, because no oxygen is transferred between the reactants.
- E)** MnO_2 is the oxidising agent and the chloride ion is the reducing agent.

Answer: E

Given: $\text{MnO}_2 + 4\text{HCl} \rightarrow \text{MnCl}_2 + \text{Cl}_2 + 2\text{H}_2\text{O}$; O is -2 and H is $+1$.

Relationship: the reduced species is the oxidising agent; the oxidised species is the reducing agent.

Step 1: Mn goes from $+4$ in MnO_2 to $+2$ in MnCl_2 , a fall of 2, so MnO_2 is reduced.

Step 2: Cl goes from -1 in HCl to 0 in Cl_2 , a rise of 1 per chlorine, so Cl^- is oxidised.

Result: MnO_2 is the oxidising agent and the chloride ion is the reducing agent.

Classic slip: no oxygen transferred, so no oxidation, misreads redox as oxygen transfer; the electron bookkeeping shows two chloride ions losing electrons while oxygen never moves.

47. Chlorine gas is bubbled into cold dilute sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$. Which of the following statements is correct?

- A)** Chlorine is oxidised only, and the hydroxide ion is the oxidising agent.
- B)** Chlorine is both oxidised and reduced, while oxygen keeps oxidation number -2 throughout.
- C)** The chloride ion is oxidised, because chlorine gas gains electrons.
- D)** Oxygen is oxidised from -2 to 0 in the water that forms.
- E)** Chlorine is reduced only, and the hydroxide ion is the reducing agent.

Answer: B

Given: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$; chlorine starts at oxidation number 0 in Cl_2 ; O is -2 and H is $+1$.

Relationship: assign chlorine's oxidation number in each product and compare it with 0 .

Step 1: in NaCl, $n(\text{Cl}) = -1$: a fall of 1, so this chlorine is reduced.

Step 2: in NaClO, $n(\text{Cl}) + (-2) = -1$, so $n(\text{Cl}) = +1$: a rise of 1, so this chlorine is oxidised.

Step 3: oxygen sits at -2 in OH^- , ClO^- and H_2O throughout, and hydrogen stays $+1$; nothing else changes.

Result: chlorine is both oxidised and reduced (a disproportionation) while oxygen keeps -2 .

Classic slip: charging the hydroxide ion with reducing or oxidising fails because no atom inside it changes oxidation number; one chlorine molecule does both jobs.

48. Sulfite ions, SO_3^{2-} , are oxidised to sulfate ions, SO_4^{2-} , in acidic solution. When this change is written as a balanced half-equation, how many electrons appear on the right-hand side?

- A) 1 B) 2 C) 3 D) 4 E) 6**

Answer: B

Given: SO_3^{2-} oxidised to SO_4^{2-} in acidic solution; O is -2 ; sulfur is the only element that changes.

Relationship: electrons released = rise in sulfur's oxidation number; balance O with H_2O and H with H^+ .

Step 1: $n(\text{S})$ in SO_3^{2-} : $n + 3(-2) = -2$, so $n(\text{S}) = +4$

Step 2: $n(\text{S})$ in SO_4^{2-} : $n + 4(-2) = -2$, so $n(\text{S}) = +6$

Step 3: rise = $6 - 4 = 2$, so 2 electrons appear on the right.

Result: $\text{SO}_3^{2-} + \text{H}_2\text{O} \rightarrow \text{SO}_4^{2-} + 2\text{H}^+ + 2\text{e}^-$, so 2 electrons; charge check: -2 on the left, and $-2 + 2 - 2 = -2$ on the right ✓.

Classic slip: 3 or 4 electrons track the oxygen atom gained instead of the two-unit climb of sulfur from $+4$ to $+6$.

49. In acidic solution, manganate(VII) ions, MnO_4^- , oxidise iron(II) ions, Fe^{2+} , to Fe^{3+} , the manganese ending as Mn^{2+} . In the balanced ionic equation, how many electrons are transferred per MnO_4^- ion, and what coefficient does Fe^{2+} carry?

- A) 3 electrons and 3 Fe^{2+}**
B) 4 electrons and 4 Fe^{2+}
C) 5 electrons and 1 Fe^{2+}
D) 5 electrons and 5 Fe^{2+}
E) 7 electrons and 7 Fe^{2+}

Answer: D

Given: $\text{MnO}_4^- \rightarrow \text{Mn}^{2+}$ and $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+}$ in acidic solution; O is -2 .

Relationship: electrons gained per MnO_4^- = fall of manganese's oxidation number; the Fe^{2+} coefficient equals that count, since each Fe^{2+} supplies 1 electron.

Step 1: $n(\text{Mn})$ in MnO_4^- : $n + 4(-2) = -1$, so $n(\text{Mn}) = +7$

Step 2: manganese falls from $+7$ to $+2$, a fall of 5, so each MnO_4^- gains 5 electrons.

Step 3: each $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+}$ loses 1 electron, so 5 Fe^{2+} are needed.

Result: 5 electrons per MnO_4^- and coefficient 5 on Fe^{2+} , giving $\text{MnO}_4^- + 8\text{H}^+ + 5\text{Fe}^{2+} \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O} + 5\text{Fe}^{3+}$ (charge $+17$ on both sides ✓).

Classic slip: 7 electrons counts manganese all the way down to 0, the metallic atom, instead of stopping at the $+2$ ion the products actually show.

Section E - Bonding, polarity and intermolecular forces (Q50-59)

50. Butane (C_4H_{10} , Mr 58), propanone ($\text{C}_3\text{H}_6\text{O}$, Mr 58) and propan-1-ol ($\text{C}_3\text{H}_8\text{O}$, Mr 60) boil at approximately 0 °C, 56 °C and 97 °C, in some order. Which assignment is correct?

- A) Butane 0 °C, propan-1-ol 56 °C, propanone 97 °C.
- B) Butane 56 °C, propanone 97 °C, propan-1-ol 0 °C.
- C) Butane 0 °C, propanone 56 °C, propan-1-ol 97 °C.
- D) Propanone 0 °C, butane 56 °C, propan-1-ol 97 °C.
- E) Propan-1-ol 0 °C, propanone 56 °C, butane 97 °C.

Answer: C

With molar masses nearly equal (58, 58, 60), boiling point is decided by the strength of intermolecular forces. Propan-1-ol hydrogen bonds through its O-H group and boils highest at 97 °C, propanone has permanent dipole-dipole forces only and boils at 56 °C, and butane has only weak London forces and boils lowest at 0 °C. Any ranking that places propanone above propan-1-ol fails because dipole-dipole attraction is weaker than hydrogen bonding at the same molar mass.

51. Butan-1-ol, butan-2-ol and 2-methylpropan-2-ol are isomers that all contain an O-H group. Their boiling points are approximately 118 °C, 100 °C and 82 °C respectively. Which statement correctly explains this trend?

- A) Branching lowers the boiling point because it weakens the O-H bonds inside the molecules.
- B) All three have the same boiling point, because hydrogen bonding dominates in each of them.
- C) The boiling points differ because the three isomers contain different functional groups.
- D) 2-methylpropan-2-ol has the highest boiling point, because its compact shape packs the molecules closest together.
- E) Butan-1-ol has the highest boiling point, because its least branched chain gives the greatest molecular surface contact and so the strongest London forces, which act in addition to the hydrogen bonding present in all three isomers.

Answer: E

All three isomers hydrogen bond equally through their O-H groups, so the boiling-point spread comes from the London force component, which grows with molecular surface contact. The unbranched chain of butan-1-ol gives the greatest contact and the highest boiling point (118 °C), while the compact branching of 2-methylpropan-2-ol gives the least contact and the lowest (82 °C). Branching alters intermolecular contact only; it cannot weaken the O-H covalent bonds inside the molecules, and all three carry the same functional group.

52. Hydrogen fluoride (Mr 20) boils at approximately 20 °C, while hydrogen chloride (Mr 36.5) boils at approximately -85 °C. Which statement explains this comparison?

- A)** HF molecules hydrogen bond to one another, a force much stronger than the dipole-dipole and London forces between HCl molecules, and this outweighs the higher molar mass of HCl.
- B)** HCl has the higher molar mass, so its London forces should make it boil higher, and the quoted temperatures cannot both be right.
- C)** The H-F covalent bond is stronger than the H-Cl covalent bond, and covalent bond strength sets the boiling point.
- D)** HF is ionic while HCl is covalent, and ionic compounds boil much higher.
- E)** HCl is non-polar, so only very weak London forces act between its molecules, while HF is also non-polar but lighter.

Answer: A

Fluorine's high electronegativity lets HF molecules form strong hydrogen bonds, while HCl molecules attract each other only by ordinary dipole-dipole and London forces, so HF boils far higher (about 20 °C versus -85 °C) despite its lower molar mass. The comparison concerns intermolecular forces, not covalent bond strength, because the H-F bonds survive boiling intact; and HCl is polar, just with a weaker dipole and no hydrogen bonding.

53. Which of the following statements is correct?

- A)** Water is non-polar, because its two identical O-H bonds pull equally in opposite directions.
- B)** Carbon dioxide is polar, because each C=O bond is strongly polar.
- C)** Boron trifluoride is non-polar, because its three identical B-F bond dipoles cancel in the trigonal planar arrangement.
- D)** Ammonia is non-polar, because the lone pair on nitrogen makes the molecule symmetrical.
- E)** Any molecule that contains polar bonds must be polar overall.

Answer: C

BF₃ is trigonal planar, and its three identical B-F dipoles at 120 degrees cancel vectorially, so the molecule is non-polar. Water is bent, so its two O-H dipoles leave a resultant and the molecule is polar; CO₂ is linear and cancels, and ammonia is pyramidal and polar. The general claim that polar bonds always make a polar molecule is disproved by every symmetric case such as BF₃ and CO₂.

54. Carbon tetrachloride, CCl₄, chloroform, CHCl₃, and dichloromethane, CH₂Cl₂, are all tetrahedral molecules. Which statement about their polarity is correct?

- A)** All three are polar, because every tetrahedral molecule containing polar bonds is polar overall.
- B)** The polarity of each molecule is decided only by its molar mass, so the heaviest molecule is the most polar.
- C)** CCl₄ is the most polar of the three, because it contains the greatest number of C-Cl dipoles.
- D)** All three are non-polar, because the tetrahedral shape of each molecule cancels every bond dipole.
- E)** CCl₄ is non-polar because its four identical C-Cl dipoles cancel, while CHCl₃ and CH₂Cl₂ are polar because the C-H bonds do not balance the C-Cl dipoles.

Answer: E

In CCl₄ the four identical C-Cl dipoles sit symmetrically in the tetrahedron and cancel exactly, so it is non-polar. Substituting Cl by H destroys the symmetry of the dipole arrangement, so CHCl₃ and CH₂Cl₂ both keep a net dipole and are polar. Adding more identical dipoles in a fully symmetric tetrahedron makes the resultant smaller, not larger, so CCl₄ is the least polar of the three, not the most.

55. Carbon dioxide and sulfur dioxide both contain polar bonds to oxygen, yet CO₂ is non-polar and SO₂ is polar. Why?

- A)** CO₂ is linear, so its two C=O bond dipoles point in opposite directions and cancel, while SO₂ is bent, so its two S=O dipoles add to give a net dipole.
- B)** Both molecules are linear, but only SO₂ has a lone pair on the central atom strong enough to create a dipole.
- C)** SO₂ is bent, which makes its S=O bonds polar, while the linear shape of CO₂ makes its C=O bonds non-polar.
- D)** SO₂ is polar because its molecules are heavier than CO₂ molecules and attract each other more strongly.
- E)** The C=O bonds in CO₂ are non-polar, because carbon and oxygen have almost equal electronegativities.

Answer: A

Both molecules contain polar bonds, so geometry decides the outcome. CO₂ is linear, its two equal C=O dipoles point in opposite directions and cancel; SO₂ is bent (the lone pair on sulfur holds the O-S-O angle near 119 degrees), so its two S=O dipoles add to a net dipole. Bond polarity depends on electronegativity difference, not on molecular shape, so the claim that bending changes whether the bonds themselves are polar is wrong.

56. Four unknown substances were tested, with these results. P: melting point 801 °C, conducts electricity only when molten or dissolved in water. Q: melting point –95 °C, never conducts electricity. R: melting point 1085 °C, conducts electricity as a solid. S: melting point 1710 °C, never conducts electricity, even when molten. Which statement correctly identifies one of the substances?

- A)** R is metallic, because conduction in the solid state requires mobile delocalised electrons.
- B)** Q is ionic, because ionic bonds are weak and break at a low temperature.
- C)** S is ionic, because its very high melting point proves strong ionic bonds.
- D)** P is metallic, because it conducts electricity when molten.
- E)** P is a simple molecular substance, because molecular substances dissolve well in water.

Answer: A

Conduction through a solid requires mobile charge carriers, and only the delocalised electrons of a metal can move through a rigid lattice, so R, which conducts as a solid, is metallic. P conducts only when molten or dissolved, the signature of an ionic solid whose ions are locked until melting or solution frees them, and S, which never conducts even when molten, is giant covalent rather than ionic, because molten ionic compounds do conduct, so a very high melting point alone cannot mark ionic bonding.

57. Sodium chloride melts at 801 °C while iodine melts at 114 °C, although iodine has the larger molar mass. Which statement explains the comparison?

- A)** Iodine melts easily because the covalent bonds between its molecules are weak.
- B)** Sodium chloride has the higher melting point because its molar mass is greater than that of iodine.
- C)** Melting sodium chloride means overcoming strong electrostatic attractions between oppositely charged ions in a giant lattice, while melting iodine only overcomes weak London forces between I₂ molecules, and the I-I covalent bonds stay intact.
- D)** Sodium chloride contains covalent bonds, which are stronger than the covalent bonds inside iodine molecules.
- E)** Both solids are giant lattices, and iodine simply has a lattice built from lighter particles.

Answer: C

Melting NaCl means overcoming the strong electrostatic attractions between Na⁺ and Cl[–] ions repeated throughout a giant lattice, which takes 801 °C. Solid iodine is simple molecular, so melting only overcomes the weak London forces between I₂ molecules while every I-I covalent bond stays intact, which takes only 114 °C. The forces between separate iodine molecules are intermolecular London forces, not covalent bonds, and molar mass cannot compare an ionic lattice with a molecular solid.

58. Use these average bond energies: H-H 436 kJ/mol, Cl-Cl 242 kJ/mol, H-Cl 431 kJ/mol. For the reaction $\text{H}_2 + \text{Cl}_2 \rightarrow 2\text{HCl}$, the enthalpy change is approximately:

- A)** -368 kJ/mol **B)** -247 kJ/mol **C)** -184 kJ/mol **D)** +184 kJ/mol **E)** +247 kJ/mol

Answer: C

Given: average bond energies H-H 436 kJ/mol, Cl-Cl 242 kJ/mol, H-Cl 431 kJ/mol; $\text{H}_2 + \text{Cl}_2 \rightarrow 2\text{HCl}$.

Relationship: $\Delta H = (\text{energy to break bonds}) - (\text{energy released forming bonds}) = \Sigma \text{ broken} - \Sigma \text{ formed}$.

Substitute: $\Sigma \text{ broken} = 436 + 242$; $\Sigma \text{ formed} = 2 \times 431$

Step 1: $\Sigma \text{ broken} = 436 + 242 = 678 \text{ kJ/mol}$

Step 2: $\Sigma \text{ formed} = 2 \times 431 = 862 \text{ kJ/mol}$

Step 3: $\Delta H = 678 - 862 = -184 \text{ kJ/mol}$

Result: $\Delta H \approx -184 \text{ kJ/mol}$, exothermic because the bonds formed are stronger in total than those broken.

Classic slip: +184 reverses the subtraction (formed minus broken); answers built from a single H-Cl bond forget that two of them form per equation as written.

59. The thermal decomposition of calcium carbonate, $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$, has $\Delta H = +178 \text{ kJ/mol}$. Which statement is correct?

- A)** Bonds formed in the products are stronger than bonds broken in the reactants, which explains the positive ΔH .
B) The products are lower in enthalpy than the reactants, so heat is released to the surroundings.
C) At a sufficiently high temperature ΔH becomes negative, so the decomposition turns exothermic.
D) The positive sign means energy is released by the system, so the reaction is exothermic.
E) The products are higher in enthalpy than the reactants, and the difference is absorbed as heat from the surroundings.

Answer: E

A positive ΔH marks an endothermic change: the products sit higher in enthalpy than the reactants and the difference is absorbed as heat from the surroundings. The claim that stronger bonds form in the products is backwards, because stronger bonds forming releases energy and would drive ΔH negative, and ΔH does not flip sign merely because the mixture is hot.

Section F - Kinetics and catalysis (Q60-65)

60. For reactions in dilute solution at a fixed temperature, the units of reaction rate are always $\text{mol}/(\text{dm}^3 \cdot \text{s})$. Which statement about rate constants is correct?

- A)** A zero-order rate constant has the units s^{-1} .
B) A first-order rate constant has the units s^{-1} , and a second-order rate constant has the units $\text{dm}^3/(\text{mol} \cdot \text{s})$.
C) A first-order rate constant has the units $\text{mol}/(\text{dm}^3 \cdot \text{s})$.
D) Every rate constant has the same units as the rate itself, $\text{mol}/(\text{dm}^3 \cdot \text{s})$, whatever the order of reaction.
E) A second-order rate constant has the units $\text{mol}/(\text{dm}^3 \cdot \text{s})$.

Answer: B

In $\text{rate} = k[\text{A}]^n$, k must carry whatever units make the right-hand side come out in $\text{mol}/(\text{dm}^3 \cdot \text{s})$. Zero order gives k the rate's own units, first order gives s^{-1} , and second order gives $\text{dm}^3/(\text{mol} \cdot \text{s})$. So the correct pairing is a first-order constant in s^{-1} with a second-order constant in $\text{dm}^3/(\text{mol} \cdot \text{s})$; attaching $\text{mol}/(\text{dm}^3 \cdot \text{s})$ to a first-order constant actually describes the zero-order case.

61. A drug is removed from the bloodstream by first-order kinetics with a half-life of 4 hours. Starting from an initial dose D , the amount still present 12 hours after administration is:

- A)** $1/16$ of D **B)** $1/8$ of D **C)** $1/6$ of D **D)** $1/4$ of D **E)** $1/2$ of D

Answer: B

Given: first-order elimination with half-life 4 hours; initial dose D ; elapsed time 12 hours.

Relationship: after each half-life the remaining amount halves; number of half-lives = elapsed time $\div$ half-life.

Substitute: number of half-lives = $12 \text{ h} \div 4 \text{ h}$

Step 1: number of half-lives = 3

Step 2: $D \rightarrow D/2$ (4 h) $\rightarrow D/4$ (8 h) $\rightarrow D/8$ (12 h), i.e. $(1/2)^3 = 1/8$

Result: $1/8$ of D remains.

Classic slip: $1/6$ divides one half by three, treating decay as a straight line; each half-life removes half of what remains, not a fixed slice of the original dose.

62. For the reaction $A + B \rightarrow \text{products}$, kept at constant temperature: doubling $[A]$ while holding $[B]$ constant doubles the rate, and doubling $[B]$ while holding $[A]$ constant increases the rate fourfold. Which statement is correct?

- A)** The reaction is second order overall, and k has the units $\text{dm}^3/(\text{mol}\cdot\text{s})$.
B) The rate equation is $\text{rate} = k[A][B]^2$, the overall order is three, and k has the units $\text{dm}^3/(\text{mol}\cdot\text{s})$.
C) The reaction is fourth order overall, because the fourfold increase in the second experiment means the exponent of B is three.
D) The rate equation is $\text{rate} = k[A][B]^2$, the overall order is three, and k has the units $\text{dm}^6/(\text{mol}^2\cdot\text{s})$.
E) The reaction is first order overall, and k has the units s^{-1} , because each reactant contributes one power.

Answer: D

Given: doubling $[A]$ at fixed $[B]$ doubles the rate; doubling $[B]$ at fixed $[A]$ quadruples the rate; rate is measured in $\text{mol}/(\text{dm}^3\cdot\text{s})$.

Relationship: $\text{rate} = k[A]^a[B]^b$; the factor by which a concentration multiplies the rate gives the exponent; units of $k = \text{rate units} \div \text{concentration}^{\text{(overall order)}}$.

Step 1: doubling $[A]$ doubles the rate, so $a = 1$

Step 2: doubling $[B]$ quadruples the rate, and $2^b = 4$, so $b = 2$

Step 3: overall order = $1 + 2 = 3$, so $\text{rate} = k[A][B]^2$

Step 4: units of $k = (\text{mol}/(\text{dm}^3\cdot\text{s})) \div (\text{mol}/\text{dm}^3)^3 = (\text{mol}/(\text{dm}^3\cdot\text{s})) \times (\text{dm}^9/\text{mol}^3) = \text{dm}^6/(\text{mol}^2\cdot\text{s})$

Result: $\text{rate} = k[A][B]^2$, overall order three, k in $\text{dm}^6/(\text{mol}^2\cdot\text{s})$.

Classic slip: $\text{dm}^3/(\text{mol}\cdot\text{s})$ belongs to a second-order constant; a third-order law divides the rate by concentration cubed, not squared.

63. The Haber process, $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ with $\Delta H = -92 \text{ kJ/mol}$, uses an iron catalyst. Which statement is correct?

- A)** The catalyst lowers the activation energy of the forward and reverse reactions equally, so equilibrium is reached sooner, but the equilibrium position and ΔH are unchanged.
- B)** The catalyst raises the activation energy of the reverse reaction, so ammonia accumulates.
- C)** The catalyst works by raising the average kinetic energy of the gas molecules, so that more collisions are effective.
- D)** The catalyst makes the reaction more exothermic, which increases the yield at the same temperature.
- E)** The catalyst shifts the equilibrium position to the right, increasing the final yield of ammonia.

Answer: A

A catalyst offers a lower-activation-energy pathway and speeds the forward and reverse reactions by the same factor, so the mixture reaches equilibrium sooner while the equilibrium position, K and ΔH all stay unchanged. The claim that it shifts equilibrium toward ammonia fails because the yield is fixed by thermodynamics, and raising the average kinetic energy of molecules is what heating does, not what a catalyst does.

64. Manganese(IV) oxide speeds up the decomposition of hydrogen peroxide, $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$, and the black powder can be recovered unchanged at the end. The best explanation is that MnO_2 :

- A)** is consumed to supply the energy that the reaction needs, which is why a little disappears each time.
- B)** raises the average kinetic energy of the H_2O_2 molecules, so more collisions exceed the activation energy.
- C)** shifts the equilibrium of the decomposition towards the products, so the reaction goes faster.
- D)** reduces the enthalpy change of the decomposition, which releases more energy to drive the reaction.
- E)** provides an alternative reaction pathway of lower activation energy and is regenerated during the cycle, so it is not consumed.

Answer: E

MnO_2 provides the decomposition with an alternative pathway of lower activation energy and is regenerated as the cycle turns, which is exactly why the same black powder can be recovered unchanged and reused. The solid is not consumed (its recovery disproves that), a catalyst does not alter the enthalpy change of the reaction, and raising molecular speeds is what heating accomplishes, not catalysis.

65. A reaction mixture is heated from 25°C to 55°C and the rate rises sharply. According to collision theory, the main reason is that:

- A)** molecules collide much more often, and this increase in collision frequency alone explains the whole rate rise.
- B)** faster molecules are more likely to meet in the correct orientation, because high speed corrects the approach angle.
- C)** the activation energy of the reaction decreases as the temperature rises.
- D)** a much larger fraction of molecules now has energy at or above the activation energy, so far more collisions are effective.
- E)** the enthalpy change of the reaction becomes more negative at the higher temperature.

Answer: D

Heating reshapes the Maxwell-Boltzmann energy distribution so that a much larger fraction of molecules has energy at or above the activation energy, and it is this surge in effective collisions that drives the sharp rate rise. The activation energy is a property of the reaction pathway and does not fall when the mixture is warmed, and collision frequency rises only slightly over a 30 degree change, far too little to account for the whole effect.